

Multimodal RAG

Satya Pattnaik

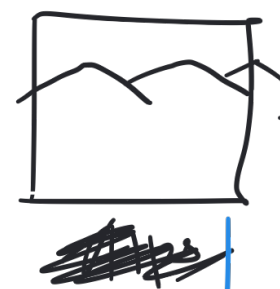
Multimodality - [Text Image] $(img \rightarrow Qwen2VL \rightarrow text)$

Text $\rightarrow GPT \rightarrow$ Text



"Swiss Alps"
↓
BERT
↓
[]

Unimodal text



Model $\rightarrow$ "text"
↓
BERT

Unimodal Image

ResNet

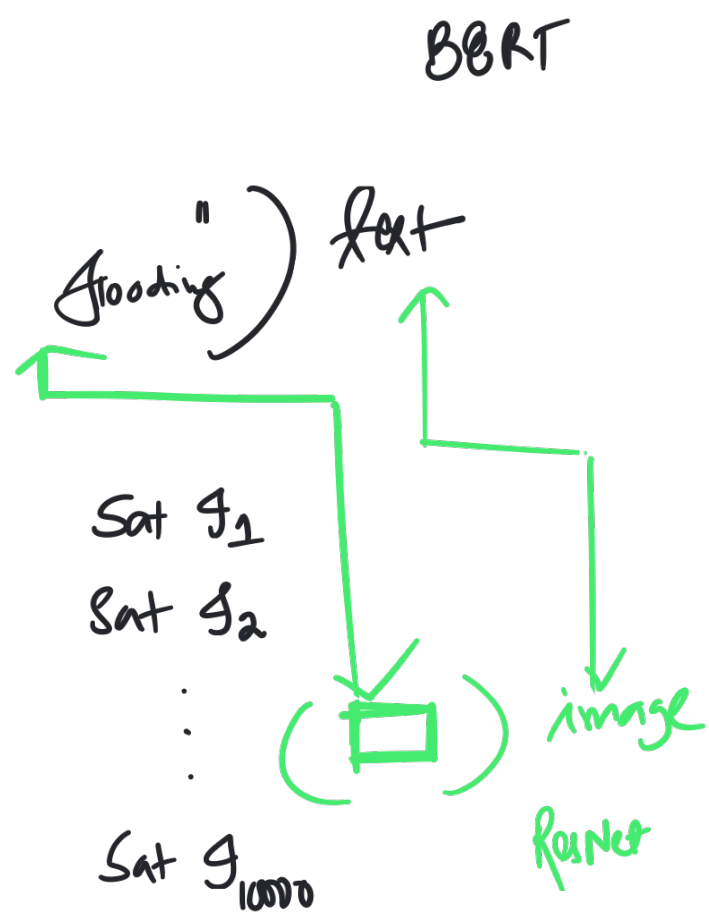
$CS = 0.9?$

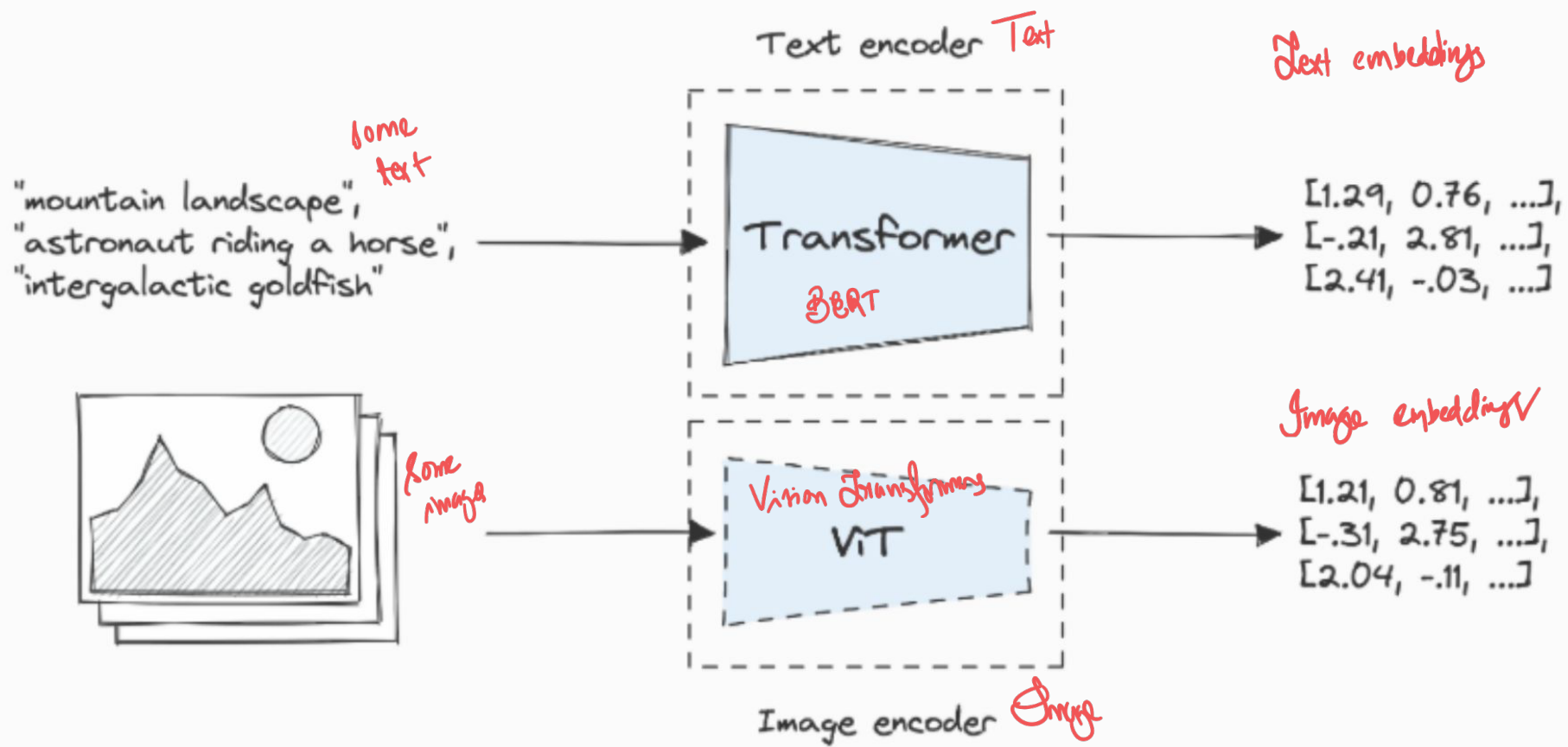
[]

Image/videos $\xrightarrow{Qwen2VL}$ text

Find satellite images showing Query ("urban")

BERT $\rightarrow \mathbf{e}$
ResNet $\rightarrow \mathbf{e}$) are not in the same space
cosine sim / any similarity will fail





we can now have $img/text$ in
the same embedding space
"an elephant in the
Serengeti"

embeddings from CLIP



text

"two dogs running across
a frosty field"

e_{text}



e_{img}



img



SBERT

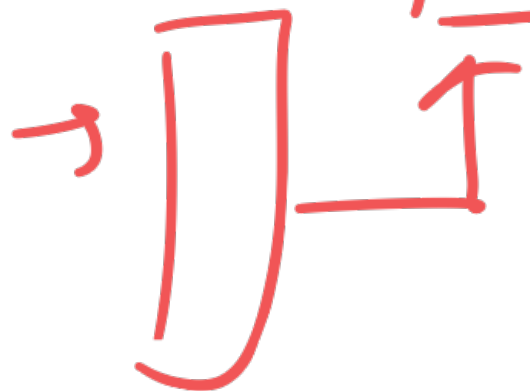
text_Q

→ BERT



text_{Doc}

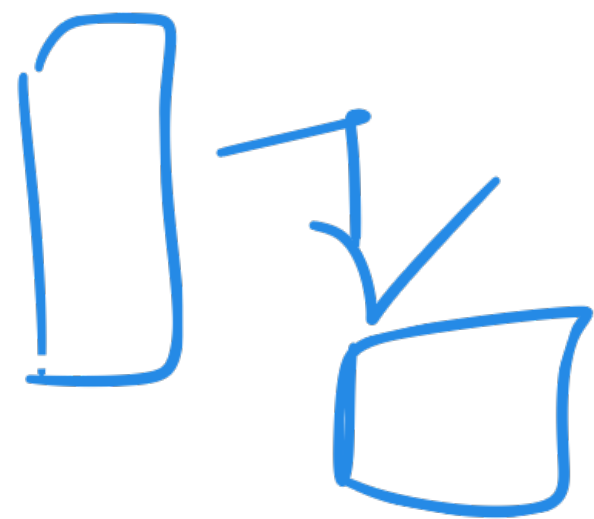
→ SBERT



clip

text

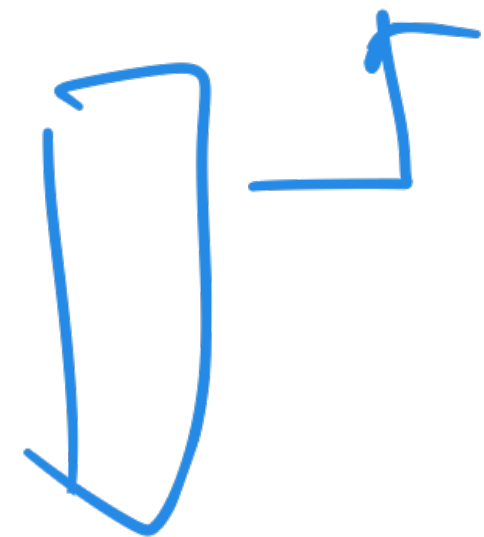
→ BERT →



img

→ ViT

→



Text

text

BERT

frozen

D_1^T

D_2^T

Emb

Sim

image

ViT

frozen

D_1^I

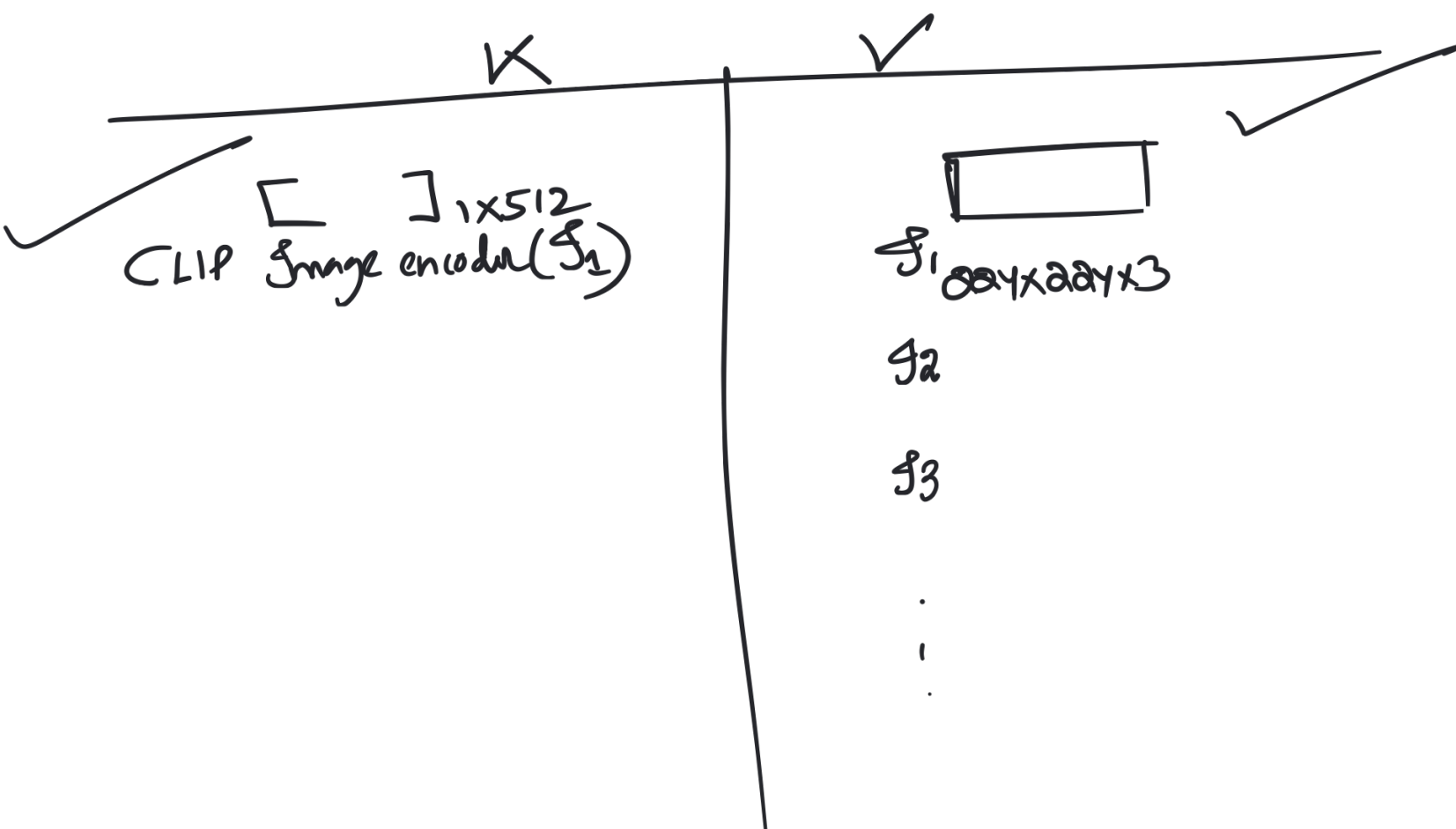
D_2^I

Emb

images

trainable

Projection Layers (trainable)



Video DB

✓
8 image

CLIP Text Encoder [text - "a sunset over mountains"]

CLIP Image Encoder (f_1)

" (f_2)

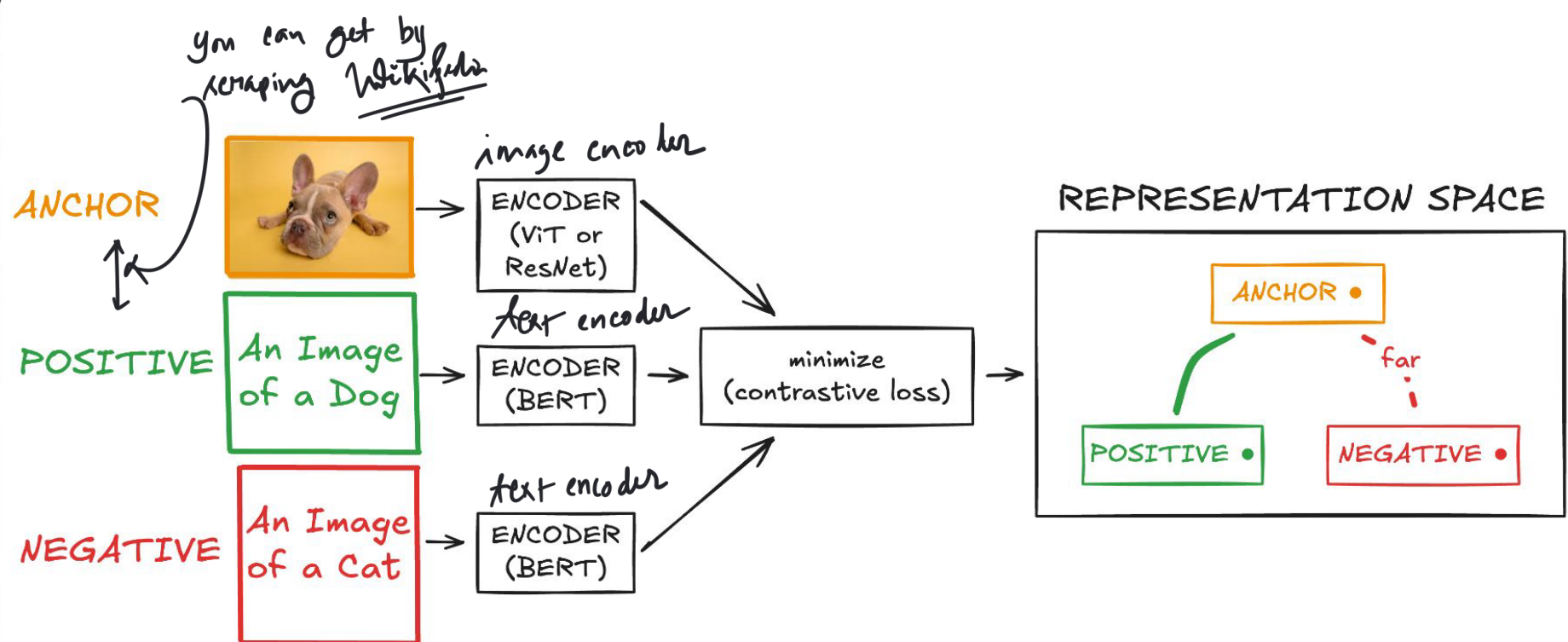
" (f_3)

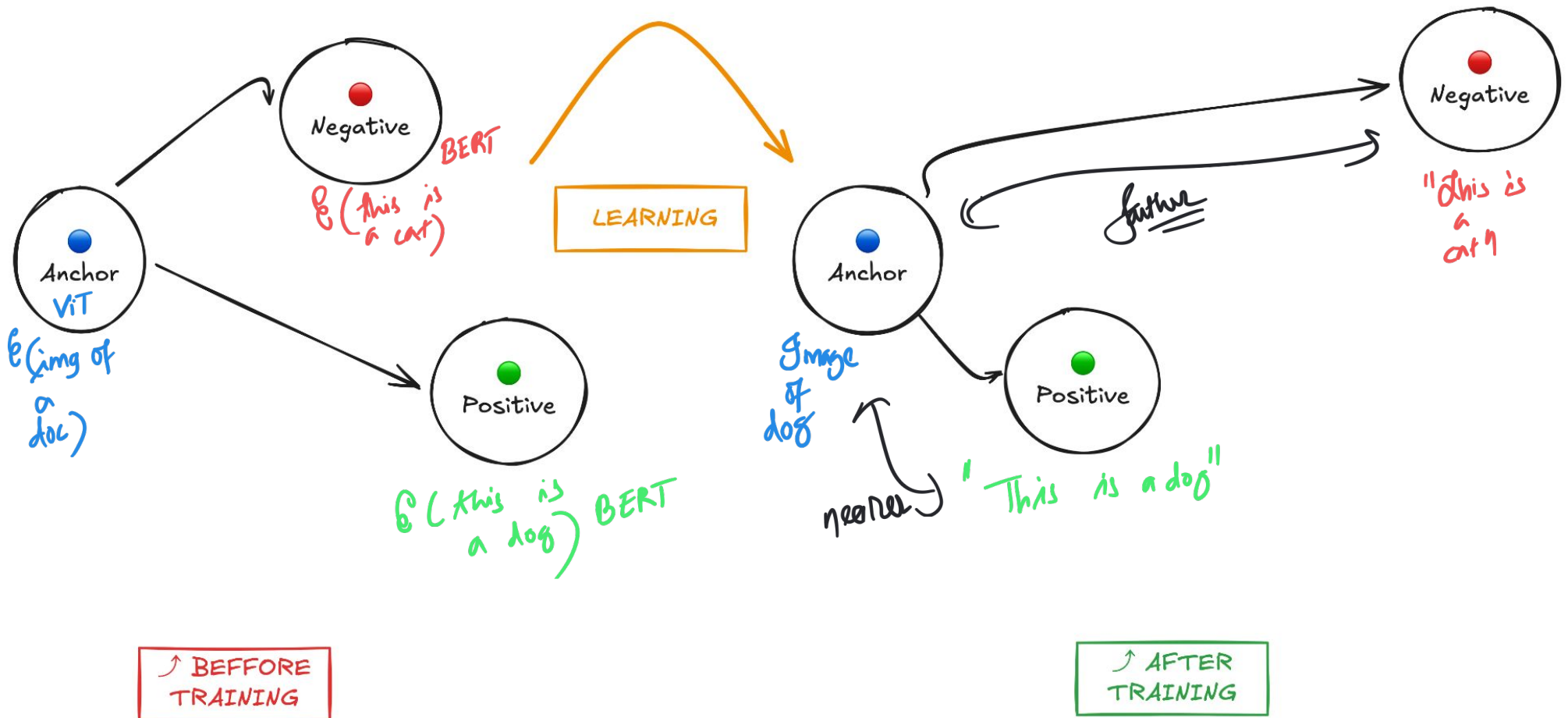
.

.

.

0.9







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Search Wikipedia

Half Dome

Page Title

From Wikipedia, the free encyclopedia

Coordinates: 37°44′46″N 119°31′59″W﻿ / ﻿37.74611°N 119.53306°W﻿ / 37.74611; -119.53306

"Half dome" redirects here. For the term in architecture, see *Semi-dome*.

Half Dome is a [granite dome](#) at the eastern end of [Yosemite Valley](#) in [Yosemite National Park](#), California. It is a well-known [rock formation](#) in the park, named for its distinct shape. One side is a sheer face while the other three sides are smooth and round, making it appear like a dome cut in half.^[3] The [granite](#) crest rises more than 4,737 ft (1,444 m) above the [valley](#) floor.

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Page Description

Image

Half Dome



Sunset over Half Dome from Glacier Point

Reference Description

Highest point

Elevation	8846 ft (2696 m) NAVD 88 ^[1]
Prominence	1,360 ft (410 m) ^[1]
Parent peak	Clouds Rest ^[1]
Coordinates	37°44′46″N 119°31′59″W﻿ / ﻿37.74611°N 119.53306°W﻿ / 37.74611; -119.53306 ^[2]

Geography



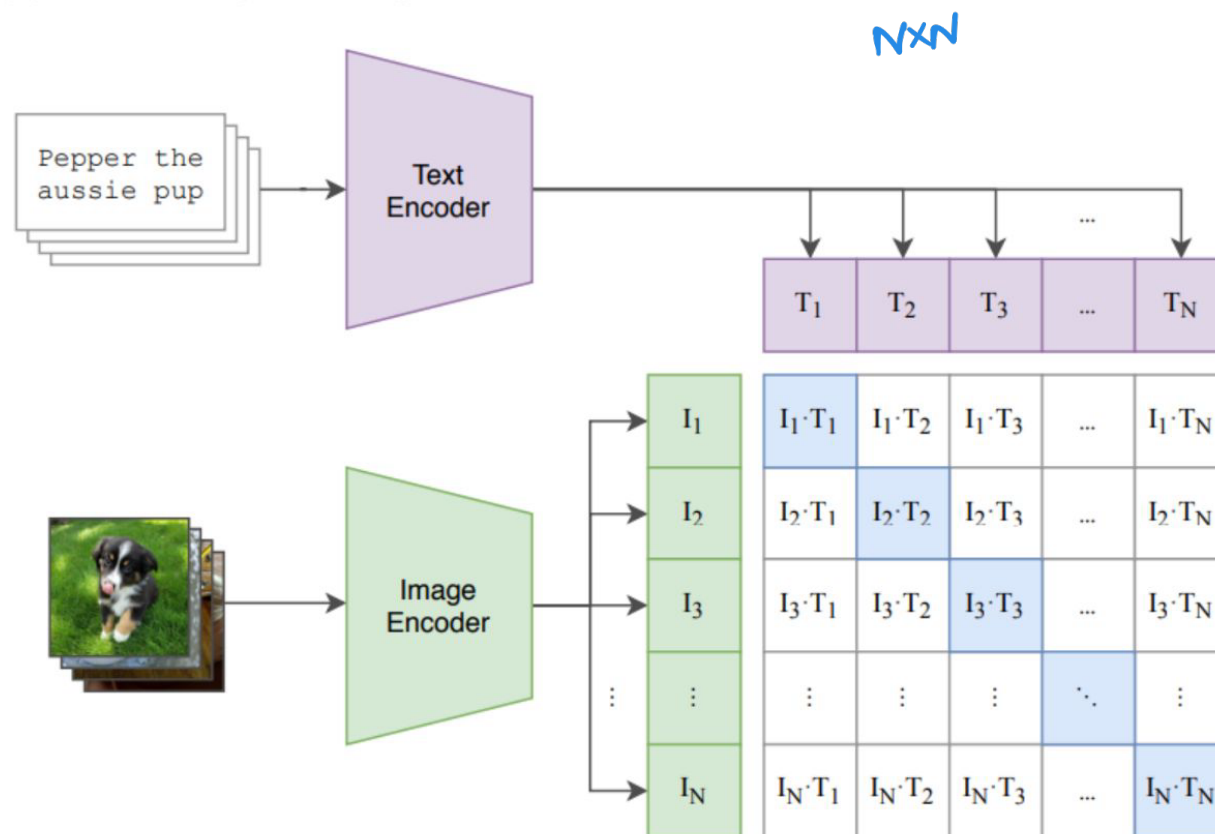
Geology

[edit]

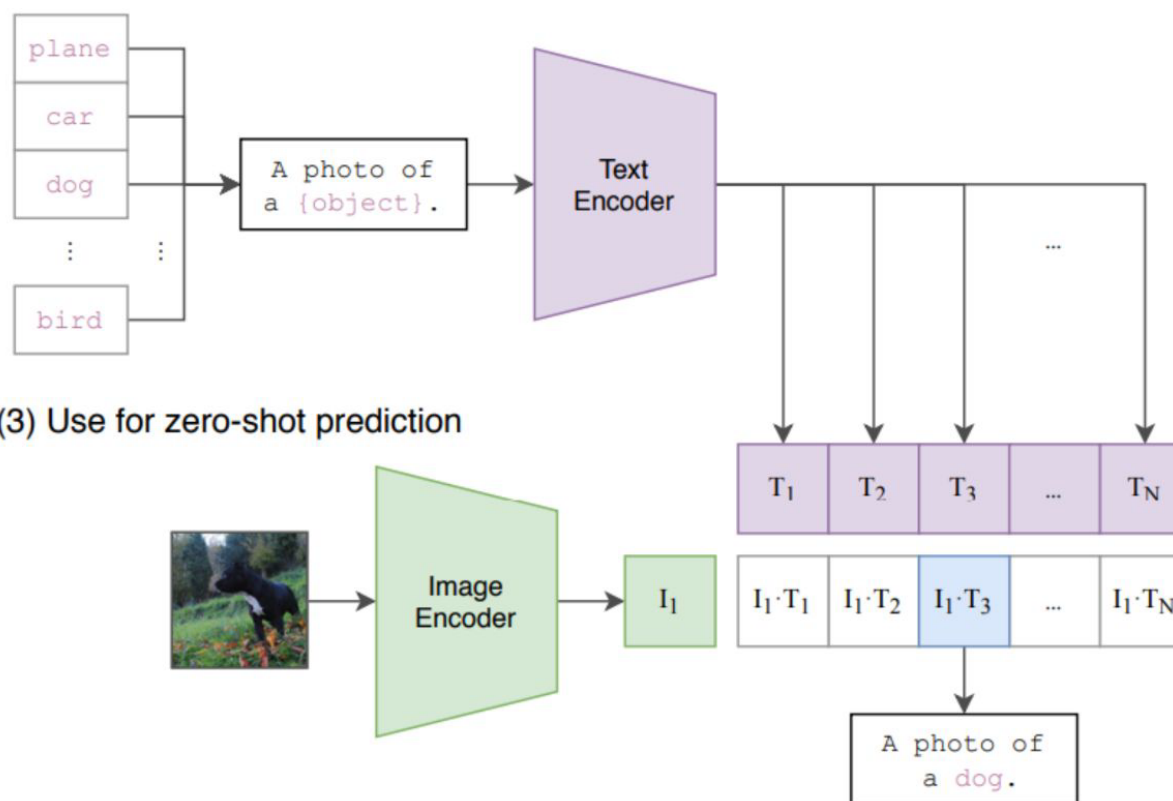
Main article: [Geology of the Yosemite area](#)

The impression from the valley floor that this is a round dome that has lost its northwest half, is just an illusion. From Washburn Point, Half Dome can be seen as a thin ridge of rock, an *arête*, that is oriented northeast-southwest, with its southeast side almost as steep as its northwest side except for the very top. Although the trend of this ridge, as well as that of *Tenaya Canyon*, is probably controlled by master *joints*, 80 percent of the northwest "half" of the original dome may well still be there.

(1) Contrastive pre-training



(2) Create dataset classifier from label text



Loss of clip

```
# image_encoder - ResNet or Vision Transformer
# text_encoder - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l] - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t - learned temperature parameter

# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i] → 3x3
T_f = text_encoder(T) #[n, d_t] → 3x3

# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1) → 3x3
T_e = l2_normalize(np.dot(T_f, W_t), axis=1) → 3x3

# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t) 9 logits

# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0) ✓
loss_t = cross_entropy_loss(logits, labels, axis=1) ✓
loss = (loss_i + loss_t)/2 ✓ 3 n's
```

img to text loss
text to img loss

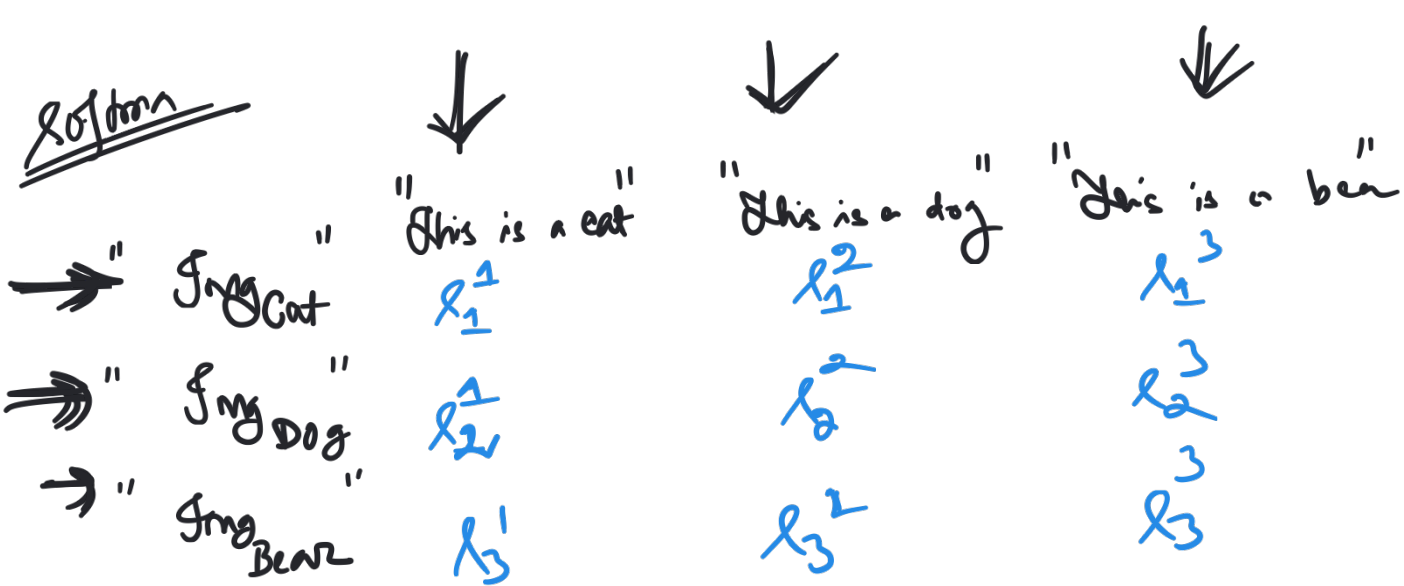


Image Text

I'm Cat	This is a cat	This is a dog	This is a bear
	<u>1</u>	0	0
I'm dog	0	1	0
I'm bear	0	0	1

text in
ring in

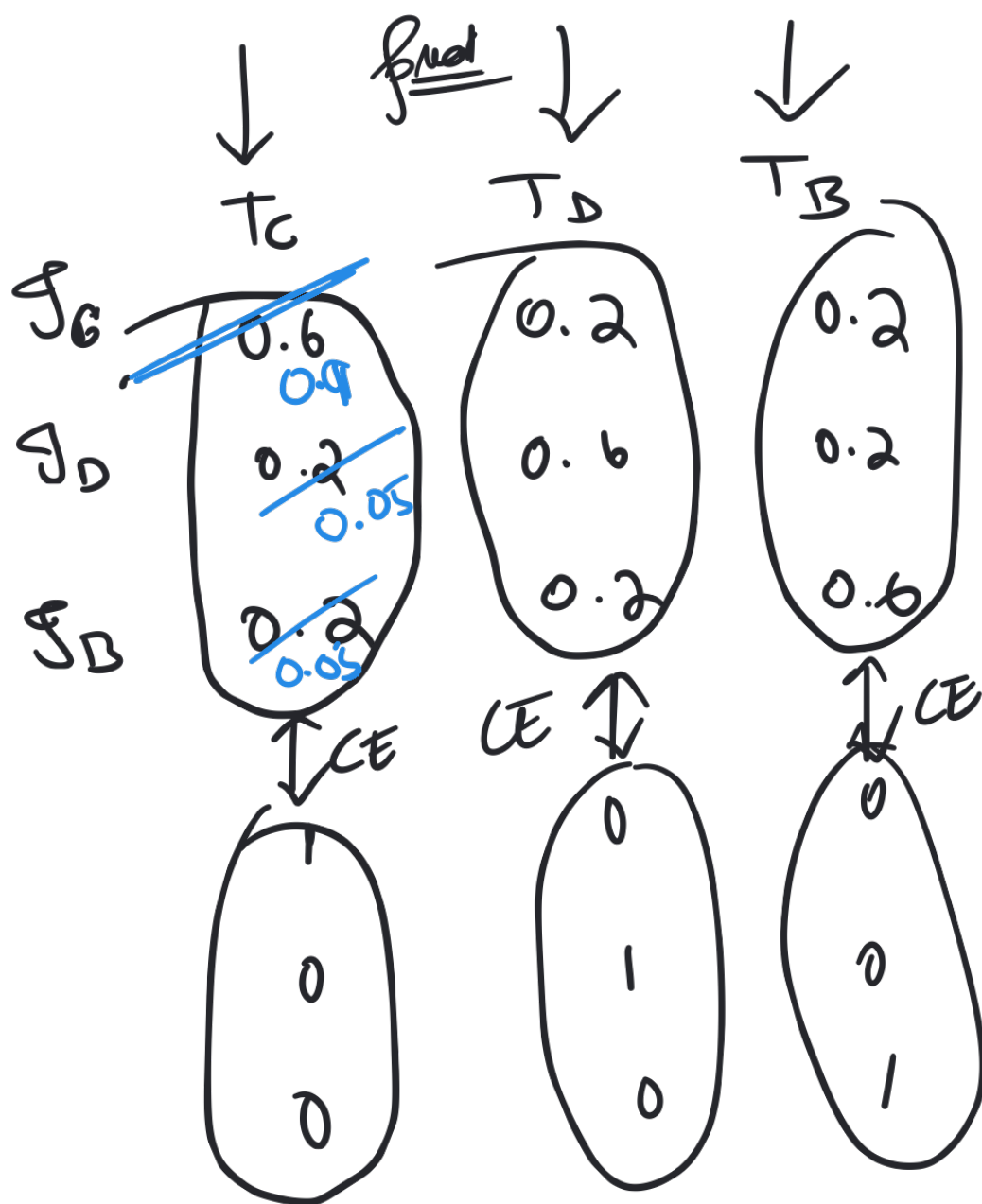
} Prediction

} Ground Truth

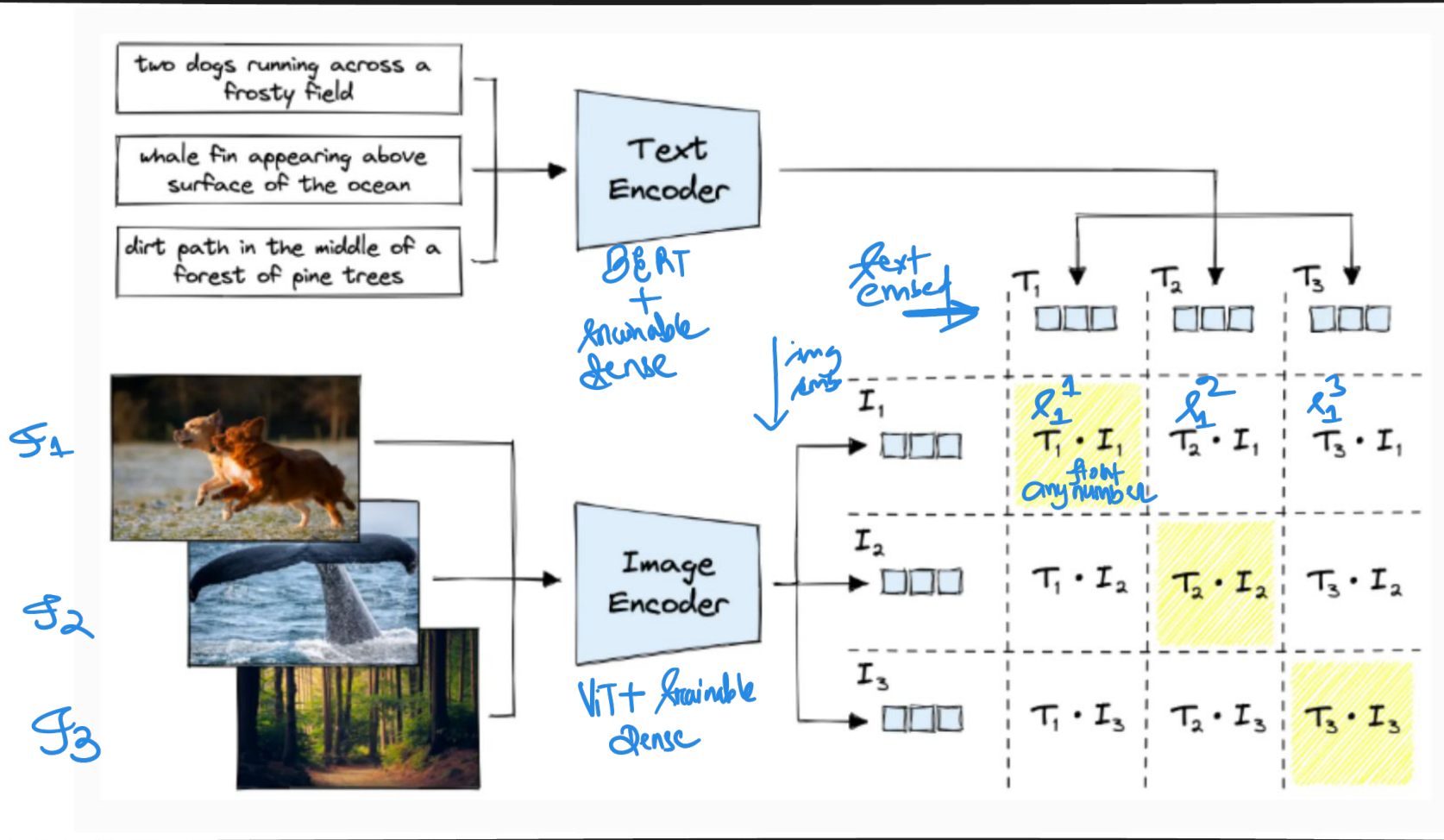
	<u>pred</u>			<u>gt</u>		
$\rightarrow s_C$	z_C	z_D	z_B			
	0.6	0.2	0.2	CE	1	0
$\rightarrow s_D$	0.7	0.15	0.15	CE	0	1
	0.2	0.6	0.2	CE	0	0
$\rightarrow s_B$	0.2	0.2	0.6			1

img - to - text loss

for



text to ing

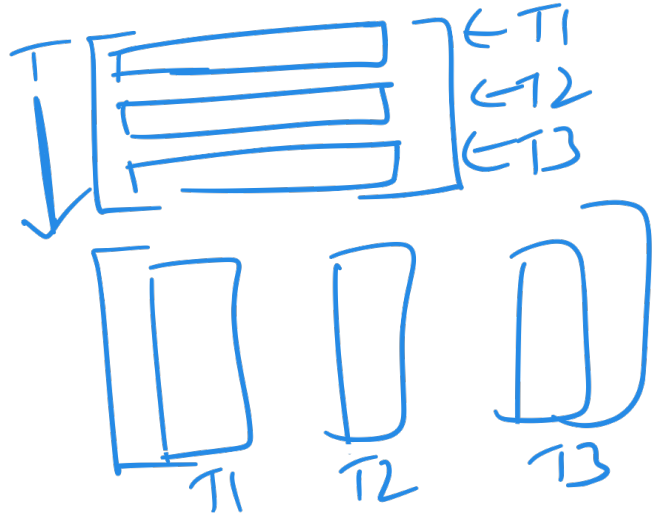
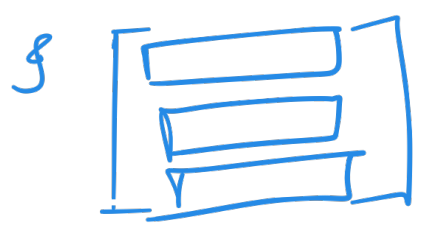


Embed size = 1x3

9:20

$I_{img} = 3 \times 3$

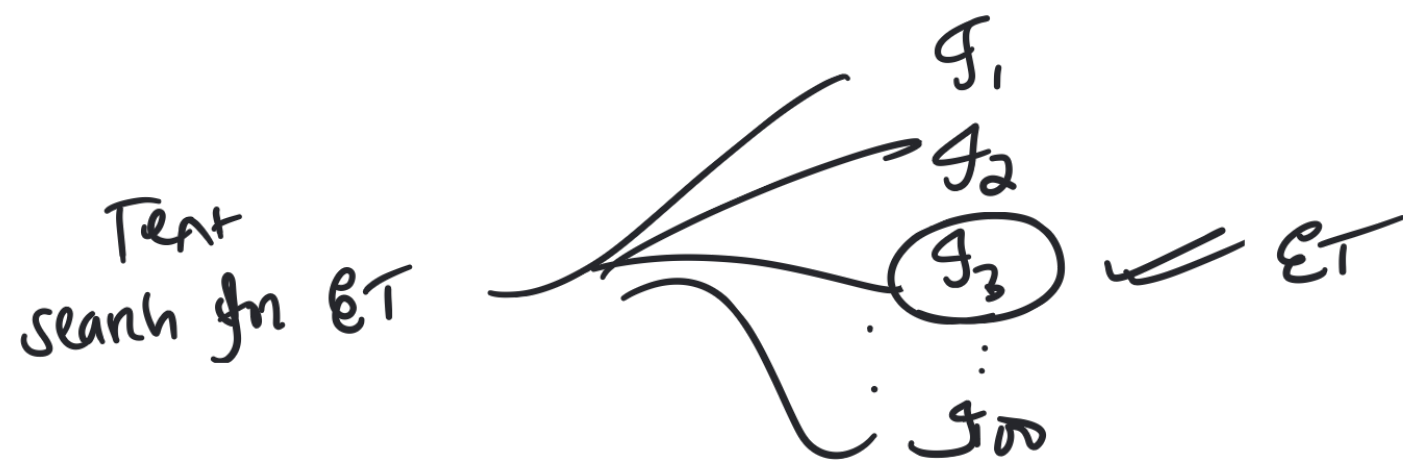
$T_{text} = 3 \times 3$



1] zero shot image classification

img $\rightarrow$ clip $\rightarrow$ class

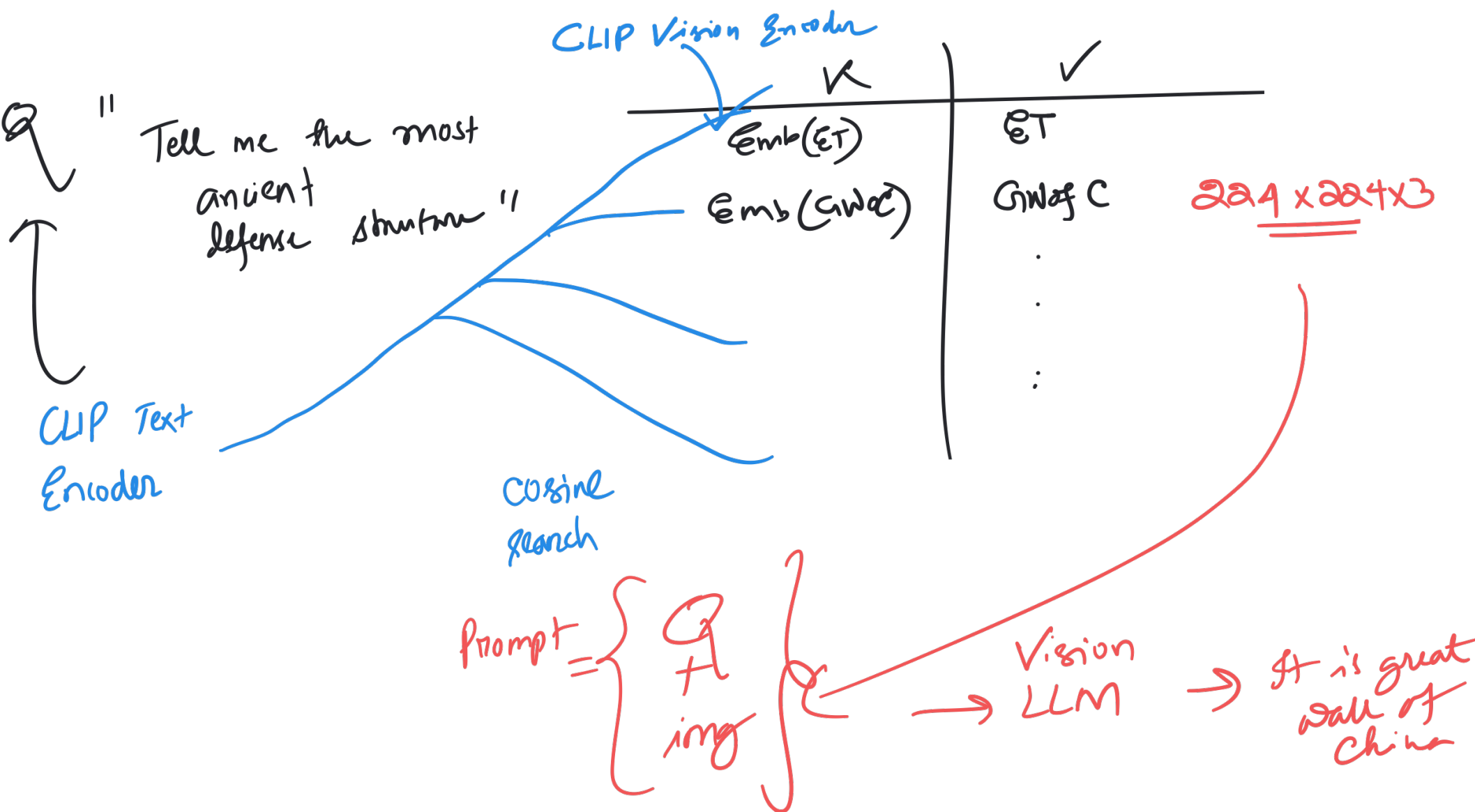
2] Text to img retrieval

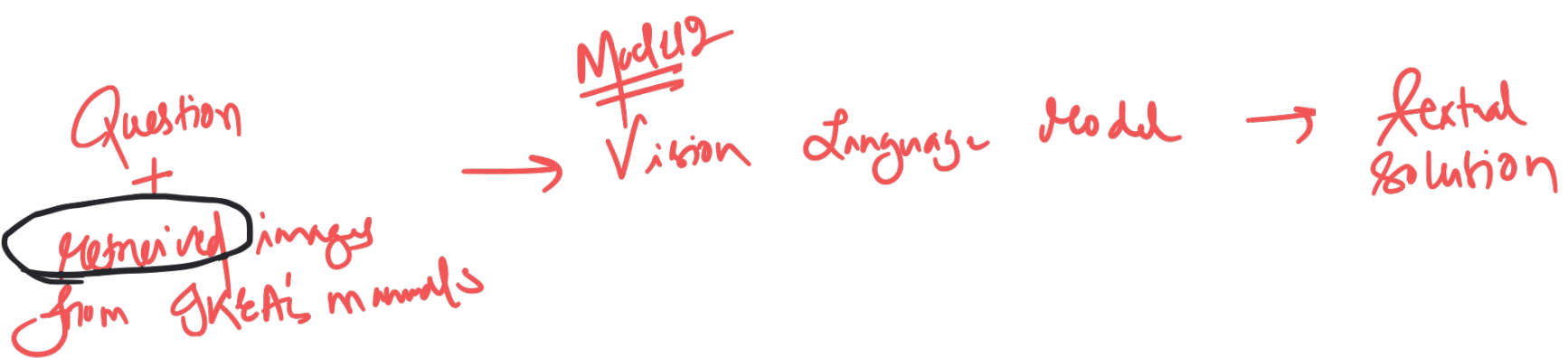


3) Image to text retrieval



4) Multimodal RAG





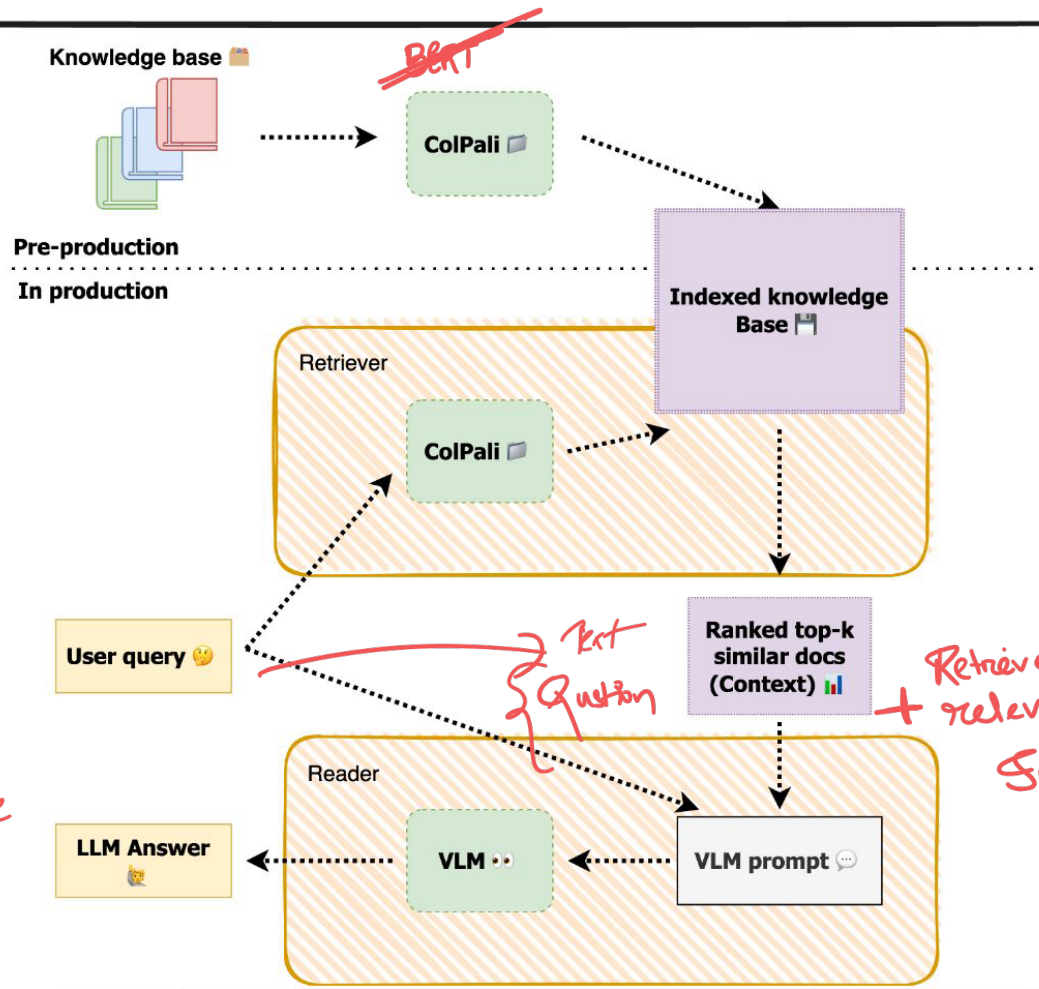


image $\rightarrow \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}$

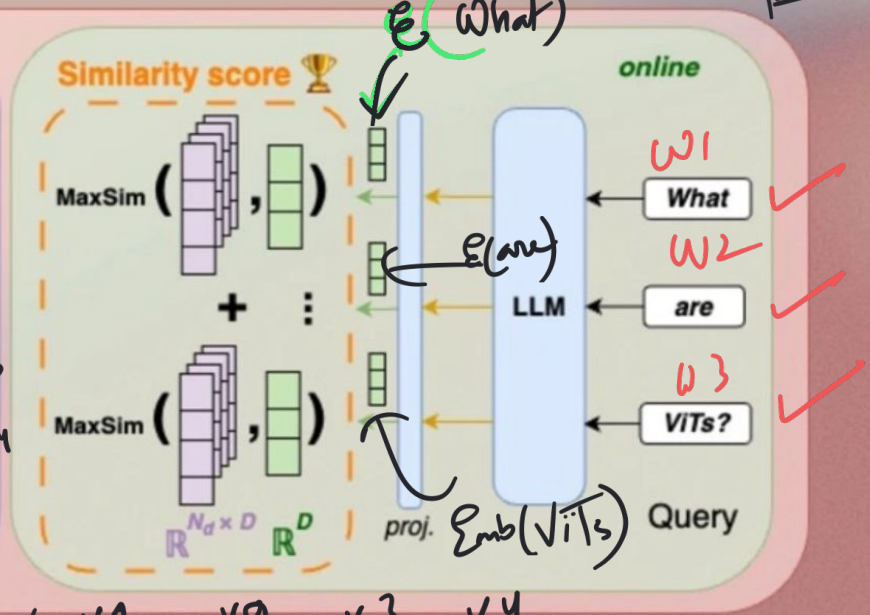
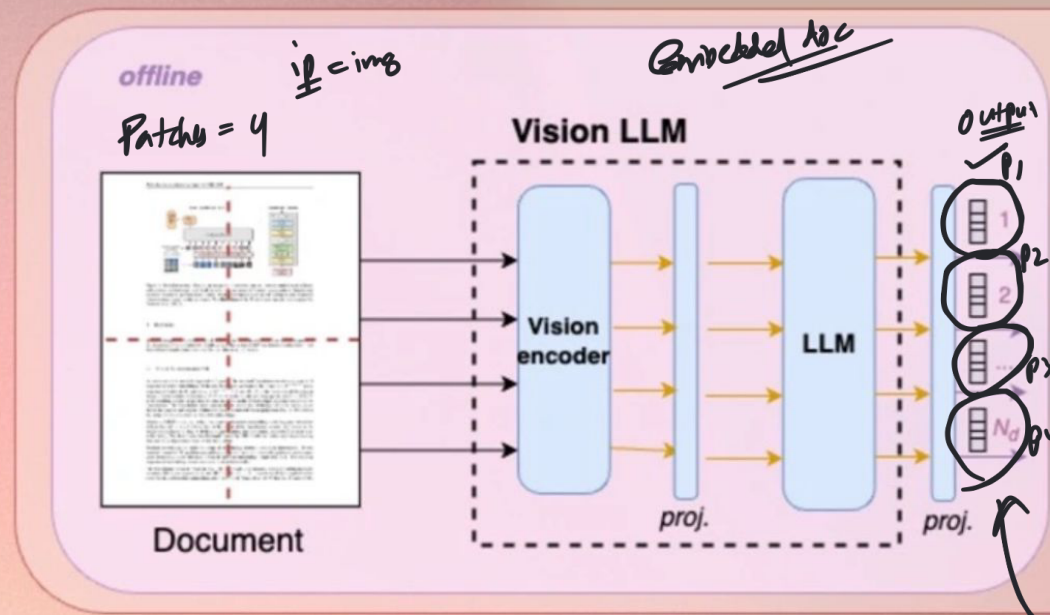
input = image of a document

* Pretrained ColPali

Query

offline
Patches = 4
if = img

Embedded doc



ColPali's Architecture

embed the info of both the images inside the doc and also text inside the doc

Doc
 $p_1 (e_1)$
 $p_2 (e_2)$
 $p_3 (e_3)$
 $p_4 (e_4)$

Doc1 = Patch1 Patch2 Patch3 Patch4

Maxim

Document $D1$	the apple is sweet and crisp
Document $D2$	the banana is ripe and yellow
Query Q	sweet apple

Document 1:

P1
P2
P3
P4

token	<u>BERT</u> embedding
"the"	[0.0, 0.0]
"apple"	[0.9, 0.1]
"is"	[0.0, 0.0]
"sweet"	[0.1, 0.9]
"and"	[0.0, 0.0]
"crisp"	[0.7, 0.7]

Document 2:

BERT

token	embedding
"the"	[0.0, 0.0]
"banana"	[0.8, 0.2]
"is"	[0.0, 0.0]
"ripe"	[0.2, 0.8]
"and"	[0.0, 0.0]
"yellow"	[0.3, 0.7]

Query:

BERT

token	embedding
"sweet"	[0.1, 0.9]
"apple"	[0.9, 0.1]

tokens	dot product
^{1st word of query} $\text{sim}(\mathbf{q}_{\text{sweet}}, \mathbf{d1}_{\text{the}})$	$[0.1, 0.9] \cdot [0.0, 0.0] = 0$
$\text{sim}(\mathbf{q}_{\text{sweet}}, \mathbf{d1}_{\text{apple}})$	$[0.1, 0.9] \cdot [0.9, 0.1] = 0.09 + 0.09 = 0.18$
$\text{sim}(\mathbf{q}_{\text{sweet}}, \mathbf{d1}_{\text{is}})$	$[0.1, 0.9] \cdot [0.0, 0.0] = 0$
$\text{sim}(\mathbf{q}_{\text{sweet}}, \mathbf{d1}_{\text{sweet}})$	$[0.1, 0.9] \cdot [0.1, 0.9] = (0.1 * 0.1) + (0.9 * 0.9) = \mathbf{0.82}$ (highest)
$\text{sim}(\mathbf{q}_{\text{sweet}}, \mathbf{d1}_{\text{and}})$	$[0.1, 0.9] \cdot [0.0, 0.0] = 0$
$\text{sim}(\mathbf{q}_{\text{sweet}}, \mathbf{d1}_{\text{crisp}})$	$[0.1, 0.9] \cdot [0.7, 0.7] = 0.07 + 0.63 = 0.70$
$\text{MaxSim}(\mathbf{q}_{\text{sweet}}, D1) = \text{max}(0.0, 0.18, 0.0, \mathbf{0.82}, 0.0, 0.70) = \mathbf{0.82}$	

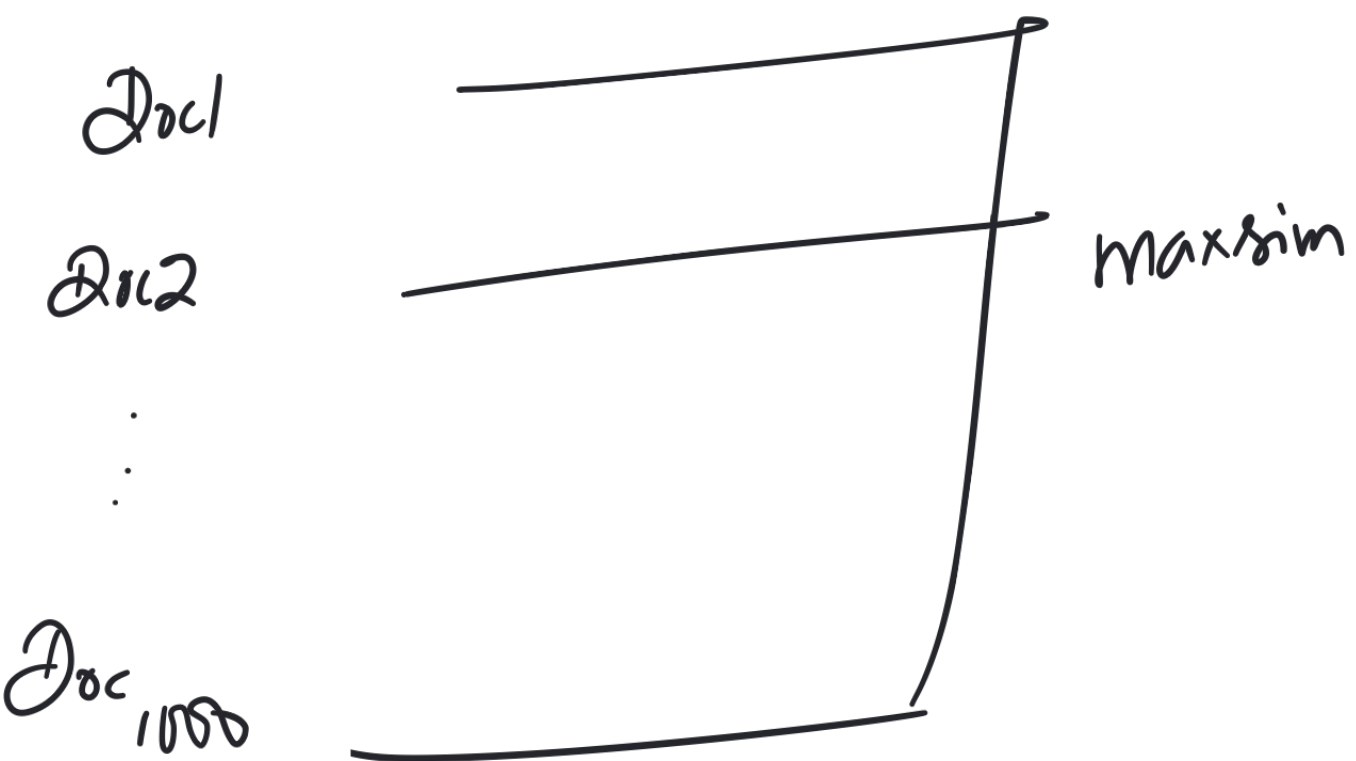
tokens	dot product
$\text{sim}(\mathbf{q}_{\text{apple}}, \mathbf{d1}_{\text{the}})$	$[0.9, 0.1] \cdot [0.0, 0.0] = 0.0$
$\text{sim}(\mathbf{q}_{\text{apple}}, \mathbf{d1}_{\text{apple}})$	$[0.9, 0.1] \cdot [0.9, 0.1] = (0.9 * 0.9) + (0.1 * 0.1) = \mathbf{0.82}$ (highest)
$\text{sim}(\mathbf{q}_{\text{apple}}, \mathbf{d1}_{\text{is}})$	$[0.9, 0.1] \cdot [0.0, 0.0] = 0.0$
$\text{sim}(\mathbf{q}_{\text{apple}}, \mathbf{d1}_{\text{sweet}})$	$[0.9, 0.1] \cdot [0.1, 0.9] = (0.9 * 0.1) + (0.1 * 0.9) = 0.18$
$\text{sim}(\mathbf{q}_{\text{apple}}, \mathbf{d1}_{\text{and}})$	$[0.9, 0.1] \cdot [0.0, 0.0] = 0.0$
$\text{sim}(\mathbf{q}_{\text{apple}}, \mathbf{d1}_{\text{crisp}})$	$[0.9, 0.1] \cdot [0.7, 0.7] = (0.9 * 0.7) + (0.1 * 0.7) = 0.63 + 0.07 = 0.70$
$\text{MaxSim}(\mathbf{q}_{\text{apple}}, D1) =$ $\text{max}(0.0, \mathbf{0.82}, 0.0, 0.18, 0.0, 0.70) = \mathbf{0.82}$	$\text{maxsim}(q, Doc1) = \underline{\underline{0.82 + 0.82}}$

Comparison with regard to Q :

document	<u>score</u>
Document 1	1.64
Document 2	1.48

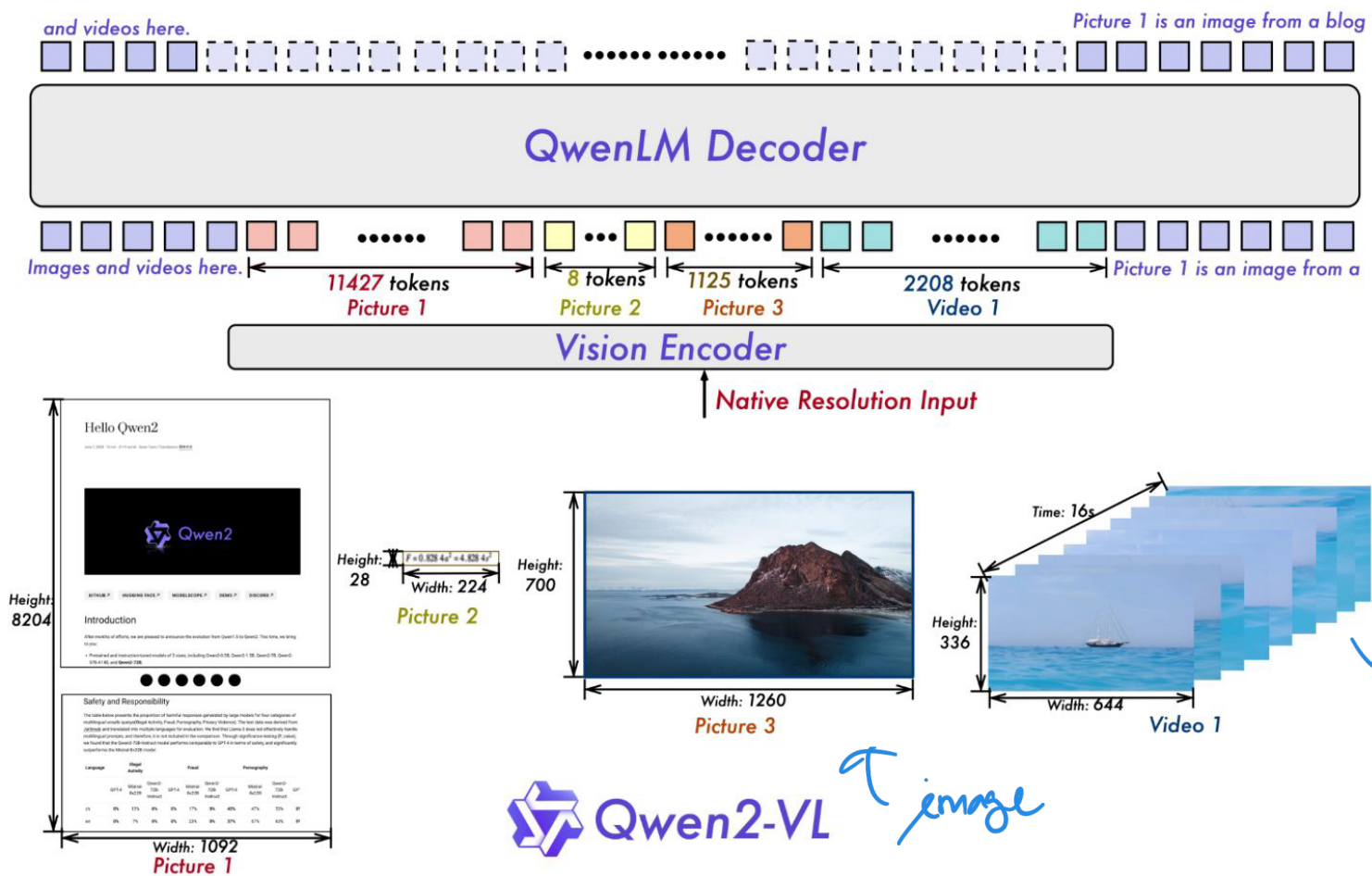
$$\text{maxsim}(q, D1) = \text{max}_{\text{word}_1 \text{ query}}(\text{word}_1, \text{Doc1}) + \text{max}_{\text{in}}(\text{word}_2, \text{Doc1})$$

Find most relevant?



$$Q = w_1 w_2$$

Textual description



good
VLM